

1 Theory: From Objective Mismatch to Error Misallocation

The experiments reveal a specific discrepancy between GPTQ and round-to-nearest (RTN). The central question is why GPTQ’s compensation can place error at behaviorally important positions.

We argue that the mechanism is an *error-allocation mismatch*. The layer-wise reconstruction objective assigns importance according to input-side activation geometry. The attention sink makes this geometry almost entirely dominated by BOS, even though the downstream loss is nearly insensitive to the BOS output and highly sensitive to template positions. GPTQ, unlike RTN, can move rounding error between columns. It therefore uses this freedom to protect the position favored by its reconstruction objective, while allowing error to increase elsewhere. The dominant-token geometry additionally predicts observable fingerprints of this redistribution: a coherent compensation direction, concentration in a small number of rows, and position-dependent damage.

Consider one linear layer $\mathbf{y}_t = \mathbf{W}\mathbf{x}_t$ with $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, where $t = 0$ denotes the BOS position and $\mathbf{x}_0$ its effective activation accumulated over calibration samples. GPTQ quantizes the input coordinates sequentially, using the input-side curvature

$$\mathbf{H} = \sum_t \mathbf{x}_t \mathbf{x}_t^\top, \quad \mathbf{H}_\lambda = \mathbf{H} + \lambda \mathbf{I}, \quad \lambda = \rho \frac{\text{tr } \mathbf{H}}{d_{\text{in}}},$$

with damping ratio ρ . We index the input coordinates in quantization order, so that act-order amounts to a permutation applied once before the loop. At step j the free set is therefore the suffix

$$F_j = \{j, j+1, \dots, d_{\text{in}}\},$$

and we work throughout on that block, writing $\mathbf{x}_{t,j} = \mathbf{x}_t[F_j]$ and, since damping is applied once to the full curvature,

$$\mathbf{H}_{\lambda,j} = \mathbf{H}_\lambda[F_j, F_j] = \mathbf{H}[F_j, F_j] + \lambda \mathbf{I}_{F_j} \succ 0.$$

Column j is rounded from $\mathbf{w}_j = \mathbf{W}_{:,j}$ to $\mathbf{q}_j$, leaving the residual $\delta_j = \mathbf{w}_j - \mathbf{q}_j \in \mathbb{R}^{d_{\text{out}}}$, and every remaining free column $k > j$ is updated by

$$\Delta \mathbf{w}_k = -\delta_j \frac{[\mathbf{H}_{\lambda,j}^{-1}]_{jk}}{[\mathbf{H}_{\lambda,j}^{-1}]_{jj}}. \quad (1)$$

Since the step changes no column outside F_j , it perturbs the output by $\Delta \mathbf{y}_t^{(j)} = \Delta \mathbf{W}^{(j)}[:, F_j] \mathbf{x}_{t,j}$, and the free set shrinks to F_{j+1} , so the next step rounds an already-modified column. RTN performs the same rounding without (1).

1.1 Objective mismatch: reconstruction protects the wrong positions

The layer-wise reconstruction objective measures how much a perturbation changes the output of the current layer, but not how costly that change is to the rest of the network. To make the missing output side explicit, consider the position-diagonal quadratic term of the downstream loss,

$$\mathcal{E}_{\text{down}}(\Delta \mathbf{W}) = \frac{1}{2} \sum_t (\Delta \mathbf{W} \mathbf{x}_t)^\top \mathbf{G}_t (\Delta \mathbf{W} \mathbf{x}_t), \quad \mathbf{G}_t = \frac{\partial^2 L}{\partial \mathbf{y}_t \partial \mathbf{y}_t^\top}. \quad (2)$$

Standard layer-wise reconstruction instead minimizes

$$\mathcal{E}_{\text{layer}}(\Delta \mathbf{W}) = \sum_t \|\Delta \mathbf{W} \mathbf{x}_t\|^2 = \text{tr} \left[\Delta \mathbf{W} \mathbf{H} \Delta \mathbf{W}^\top \right], \quad \mathbf{H} = \sum_t \mathbf{x}_t \mathbf{x}_t^\top. \quad (3)$$

Relative to (2), this objective uses the same identity metric for the output error at every position. It therefore retains the input-side geometry $\mathbf{x}_t \mathbf{x}_t^\top$ but contains no position-dependent measure of how strongly an error at position t affects the downstream loss.

This gives two position-wise quantities to compare. The input-side curvature mass of position t is

$$a_t = \|\mathbf{x}_t\|^2,$$

while we measure its downstream sensitivity by the empirical-Fisher scalar

$$g_t = \left\| \frac{\partial L}{\partial \mathbf{y}_t} \right\|^2. \quad (4)$$

Here g_t is a measurable sensitivity proxy rather than the exact Hessian block $\mathbf{G}_t$.

Proposition 1 (Position mismatch). *At the sink-forming down-projection, the BOS position carries almost all of the input-side curvature,*

$$A_{\text{BOS}} = \frac{\|\mathbf{x}_0\|^2}{\sum_t \|\mathbf{x}_t\|^2},$$

with measured values 0.998 for Llama and approximately 1.000 for Qwen, but almost none of the measured downstream sensitivity,

$$G_{\text{BOS}} = \frac{g_0}{\sum_t g_t} \leq 0.008.$$

The downstream sensitivity instead concentrates on the early template positions.

Thus the attention sink exposes a severe mismatch:

$$\boxed{A_{\text{BOS}} \approx 1 \quad \text{while} \quad G_{\text{BOS}} \approx 0.}$$

The layer-wise objective therefore concentrates its curvature on a position with little measured downstream sensitivity. This mismatch becomes potentially harmful in GPTQ because, unlike RTN, GPTQ can redistribute rounding error to reduce the layer-wise objective. We analyze this redistribution next.

1.2 Compensation turns objective mismatch into error misallocation

The distinction between GPTQ and RTN is clearest after one rounding error has been committed. RTN leaves the error where it occurs. GPTQ instead changes the remaining free columns $k > j$ to reduce (3). Consequently, GPTQ can improve its reconstruction objective while increasing error at individual positions.

At the sink-forming layer, split the free-block curvature of step j into its BOS part and the rest,

$$\mathbf{H}_{\lambda,j} = \mathbf{x}_{0,j} \mathbf{x}_{0,j}^\top + \mathbf{M}_j, \quad \mathbf{M}_j = \sum_{t \neq 0} \mathbf{x}_{t,j} \mathbf{x}_{t,j}^\top + \lambda \mathbf{I}_{F_j} \succ 0, \quad (5)$$

and define

$$\mathbf{v} = \mathbf{M}_j^{-1} \mathbf{x}_{0,j}, \quad \kappa = \mathbf{x}_{0,j}^\top \mathbf{M}_j^{-1} \mathbf{x}_{0,j}, \quad m_{kl} = [\mathbf{M}_j^{-1}]_{kl}. \quad (6)$$

The step index is fixed throughout this subsection and suppressed on $\mathbf{v}$, κ and m_{kl} ; all indices run over F_j .

Theorem 1 (Error redistribution under a dominant token). *Fix a step j : column j is rounded while the columns $k > j$ remain free.*

(a) *The inverse curvature is exactly*

$$\mathbf{H}_{\lambda,j}^{-1} = \mathbf{M}_j^{-1} - \frac{\mathbf{v}\mathbf{v}^\top}{1 + \kappa}. \quad (7)$$

Consequently, the GPTQ compensation of column $k > j$ is

$$\Delta \mathbf{w}_k = \delta_j \frac{v_j v_k / (1 + \kappa) - m_{jk}}{m_{jj} - v_j^2 / (1 + \kappa)}. \quad (8)$$

(b) *Including the rounding of column j , the output perturbation at position t is*

$$\Delta \mathbf{y}_t = -\delta_j \frac{[\mathbf{M}^{-1} \mathbf{x}_t]_j - v_j \mathbf{v}^\top \mathbf{x}_t / (1 + \kappa)}{m_{jj} - v_j^2 / (1 + \kappa)}. \quad (9)$$

At BOS,

$$\Delta \mathbf{y}_0 = -\delta_j \frac{v_j}{(1 + \kappa)m_{jj} - v_j^2}. \quad (10)$$

(c) *In the isotropic ordinary-curvature case $\mathbf{M} = \lambda \mathbf{I}$,*

$$\Delta \mathbf{w}_k = \delta_j \frac{x_{0j} x_{0k}}{\lambda + \|\mathbf{x}_{0,-j}\|^2}, \quad (11)$$

and

$$\Delta \mathbf{y}_t = -\delta_j x_{tj} + \delta_j x_{0j} \frac{\mathbf{x}_0^\top \mathbf{x}_t - x_{0j} x_{tj}}{\lambda + \|\mathbf{x}_{0,-j}\|^2}. \quad (12)$$

In particular,

$$\Delta \mathbf{y}_0 = -\delta_j x_{0j} \frac{\lambda}{\lambda + \|\mathbf{x}_{0,-j}\|^2}. \quad (13)$$

The theorem gives the central mechanism. The first term in (12),

$$-\delta_j x_{tj},$$

is the local rounding error that would also be present without compensation. The second term is introduced by GPTQ’s attempt to cancel that error under the reconstruction geometry.

At BOS, (13) shows that compensation can suppress the rounding error by a factor

$$\frac{1}{1 + \kappa_{-j}}, \quad \kappa_{-j} = \frac{\|\mathbf{x}_{0,-j}\|^2}{\lambda}.$$

The reconstruction objective therefore buys a large reduction of error at its dominant position by redistributing the committed rounding error to the remaining coordinates.

Nothing in the layer-wise objective requires this redistribution to reduce the error at every other position. When input-side importance and downstream sensitivity disagree, GPTQ can therefore improve (3) while increasing error at behaviorally sensitive positions. This is the sense in which compensation converts objective mismatch into error misallocation.

1.3 Why the empirical interventions work

The error-redistribution view also unifies the interventions that repair the failure. They act at different points in the same mechanism.

Intervention	Effect on the mechanism
Larger damping	Weakens the compensation response by increasing the regularized ordinary curvature.
Token normalization	Removes the ability of BOS to dominate the reconstruction objective solely through its activation magnitude.
Chat-formatted calibration	Adds calibration geometry associated with the positions encountered at deployment, changing where compensation can cheaply place error.
Long calibration windows	Enrich the ordinary-token curvature and reduce the relative dominance of the BOS direction.
Rotation	Spreads the concentrated sink pattern across coordinates and weakens the large coordinate-wise compensation lever.
Local RTN / re-rounding	Removes the compensation step at the culprit matrix and therefore removes the GPTQ-specific error relocation there.

These interventions need not succeed for exactly the same reason, and none alone establishes a collapse criterion. Their common implication is that the failure depends on how the reconstruction objective and compensation rule allocate error, rather than simply on the magnitude of independent rounding noise.

1.4 Restoring downstream sensitivity

The diagnosis suggests correcting the notion of token importance directly. A position-independent output-channel factor does not solve the problem.

Proposition 2 (Token weights change the allocation; common channel factors do not). *(a) Let $\mathbf{G} \succ 0$ be independent of position. Replacing the layer-wise objective by*

$$\text{tr}[\mathbf{G}\Delta\mathbf{W}\mathbf{H}\Delta\mathbf{W}^\top]$$

leaves the optimal column-wise OBS compensation rule unchanged. If $\mathbf{G}$ is diagonal, it also leaves independent nearest-grid rounding within each row unchanged.

(b) Define

$$\widetilde{\mathbf{H}} = \sum_t \frac{g_t}{\|\mathbf{x}_t\|^2} \mathbf{x}_t \mathbf{x}_t^\top. \quad (14)$$

Then

$$\text{tr} \widetilde{\mathbf{H}} = \sum_t g_t, \quad (15)$$

so the BOS share of the weighted trace is

$$\frac{g_0}{\sum_t g_t} = G_{\text{BOS}},$$

rather than the input-energy share A_{BOS} .

(c) The corresponding quadratic objective is

$$\text{tr}[\Delta \mathbf{W} \widetilde{\mathbf{H}} \Delta \mathbf{W}^\top] = \sum_t g_t \left\| \Delta \mathbf{W} \frac{\mathbf{x}_t}{\|\mathbf{x}_t\|} \right\|^2. \quad (16)$$

The two factors in (14) play distinct roles. The normalization

$$\frac{1}{\|\mathbf{x}_t\|^2}$$

removes raw activation-energy dominance, while

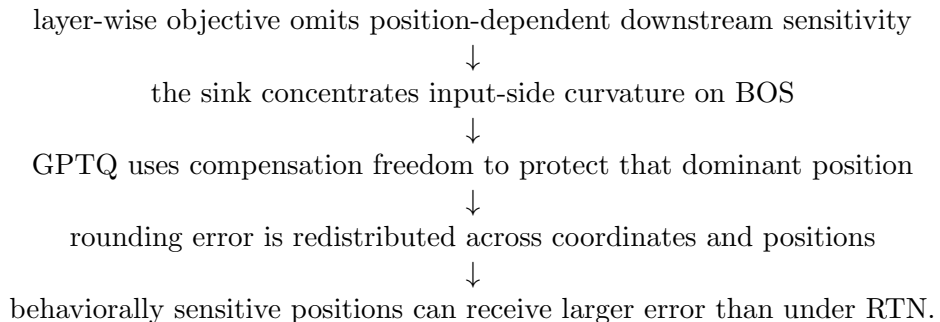
$$g_t$$

restores measured downstream sensitivity. We therefore interpret $\widetilde{\mathbf{H}}$ as a *sensitivity-aligned surrogate*, not as the exact downstream Hessian. The normalization is a modelling choice, whereas g_t is motivated by the empirical-Fisher measure of output-side sensitivity.

The corrected objective directly targets the mismatch identified in proposition 1: instead of allowing the massive BOS activation to determine which errors are cheap, it allocates curvature according to measured downstream sensitivity after normalizing away raw input energy.

1.5 Scope: lesion generation is not collapse amplification

The theory explains a local failure mechanism:



The directional and low-rank signatures are predictions of this redistribution mechanism, not the mechanism itself.

The theory also does not claim that this local error misallocation is sufficient for catastrophic instruction-following failure. The curvature mismatch occurs broadly across models with attention sinks, whereas catastrophic collapse is rare. Likewise, quantizing the sink matrix alone can increase template-position error without making GPTQ worse than RTN at the model level. Collapse therefore requires an additional interaction between the local lesion and the surrounding quantized network.

The present theory explains the generation and geometry of the local lesion and motivates its repair. The mechanism by which some quantized backgrounds amplify that lesion into model-level collapse is a separate open problem.